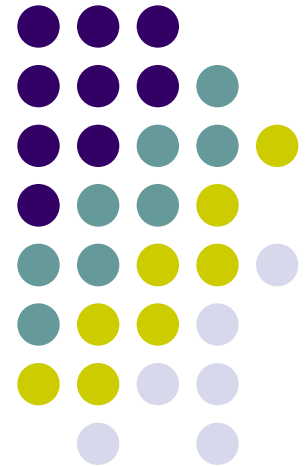


Summary

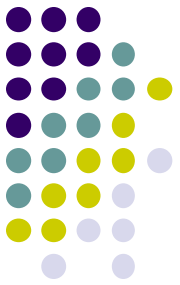
Dynamic Modelling for
Human-centered Systems



Why?

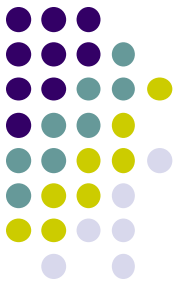


- Processes in the real world can be described and simulated by computers
- One way of embedding intelligence in systems is using computer models that
 - “understand” human behaviour and
 - can mimic human behaviour
- In this way, we can create:
 - systems that ‘understands’ desires
 - systems that provides the right support
- We do so by:
 - using knowledge from human-oriented disciplines
 - formalising this in a computational format?



What?

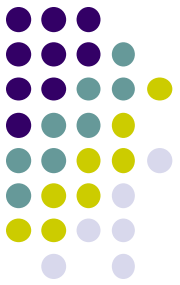
- We have learned how to make dynamic computer models of processes
 - using Excel
 - using L2-Python
- We have learned how to use models to let intelligent systems reason about humans
 - to derive conclusion about the situations
 - analysis model
 - to determine what actions to take
 - support model



Learning goals

After completing this course, the student will be able to:

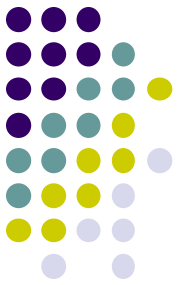
- explain the purpose of modelling and be able to describe the use of models in studying human subjects and their environment;
- describe how models are constructed, and build simple models;
- use Excel and L2-Python to study the behaviour of numerical models;
- verify that the actual behaviour of particular quantitative and qualitative models corresponds to the expected behaviour for these models; and
- report in a structured way about the process of constructing models and conducting experiments using models.
- explain how dynamic models form the basis for intelligent systems
- translate domain models into analysis models and support models



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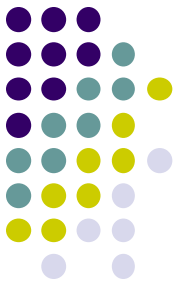
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Model

- Our focus was on **computational dynamic models**:
 - describes a specific phenomenon, theory or system (i.e. a process)
 - can simulate the process
 - can be used to study the process
- Important observations
 - a model is a *simplification*
 - during the design of the models, *choices* have been made
 - assumptions
 - a model is not the same as reality!

Users of models



- People:

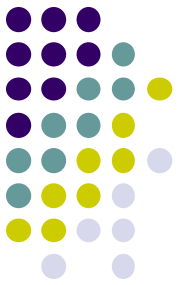
- researchers
- policy makers



- Systems:

- adaptive systems
- human-like systems



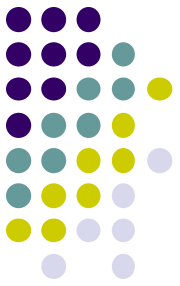


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Use cases of model validation

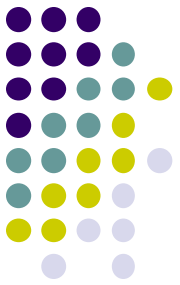


- During construction:
 - test whether model fits expectations / reality
 - “characteristic patterns”
 - result: is the model sufficiently correct
- When using for understanding and studying a phenomenon
 - scientific method: formulate hypothesis and test it
 - result: better insight in process
- Both result in “properties of model”

Properties of behaviour of dynamic models



- Follow from hypothesis / research question / expected behaviour
- Statements about:
 - effects that happen over time
 - combination of inputs that lead to some output
- Examples:
 - *Jupiter and Pluto will not collide within two years*
 - *if the interest decreases, the long term effect on household income is positive*
 - *when taking a pill every 3 hours, the medication level is between 12 and 20 mg/l*



Validation of properties

- Run (several) simulations of the model and check the properties $\Rightarrow$ scenarios
 - choose conditions / input parameters intelligently
 - be precise about what you can conclude

when is one simulation sufficient and when do you need to do several?

- Alternatively:
 - use logic / mathematics to (dis)prove the properties
 - usually quite complex

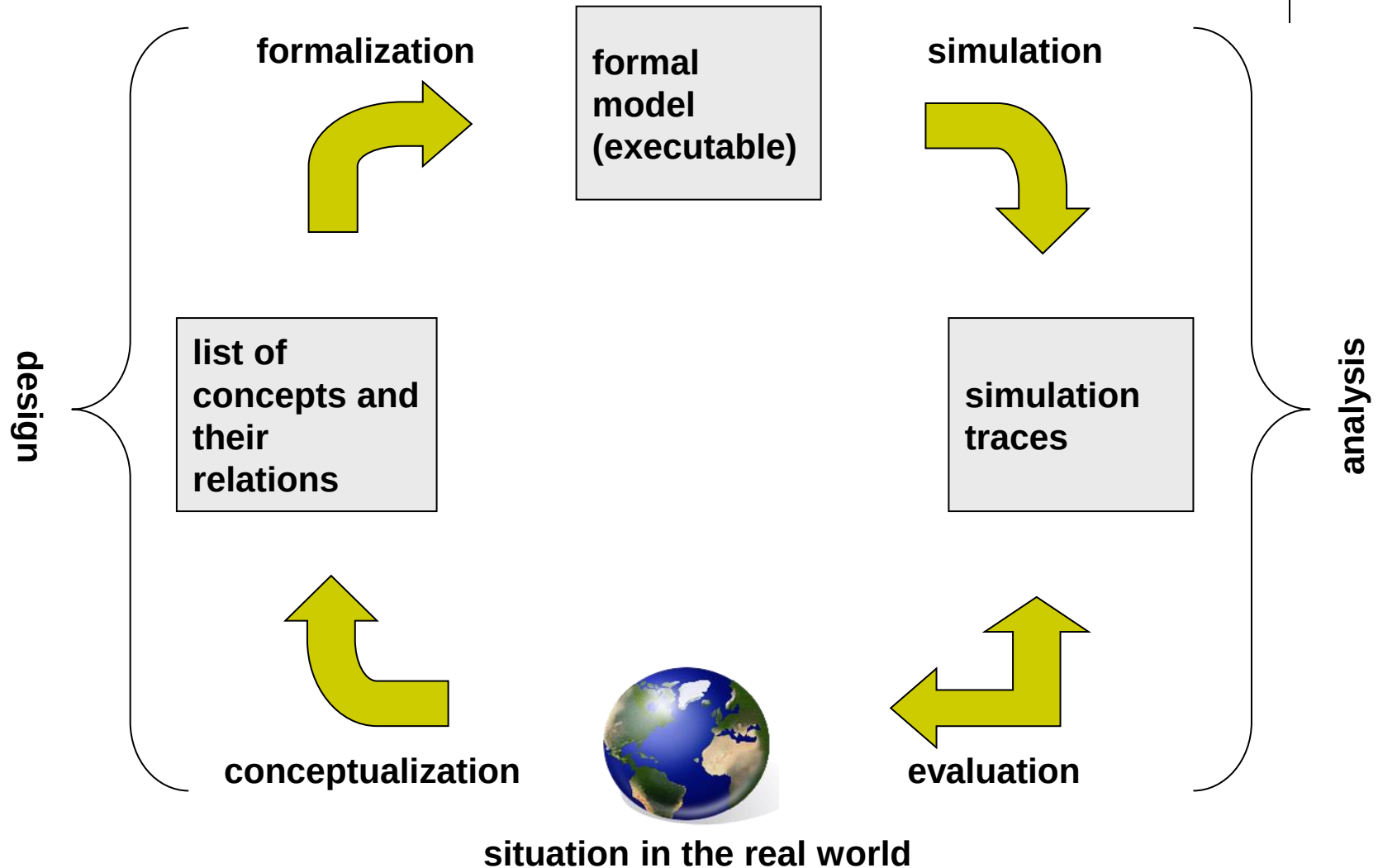
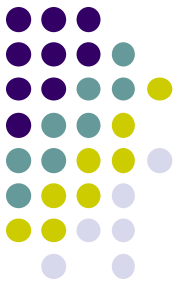


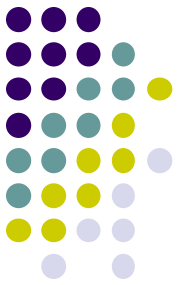
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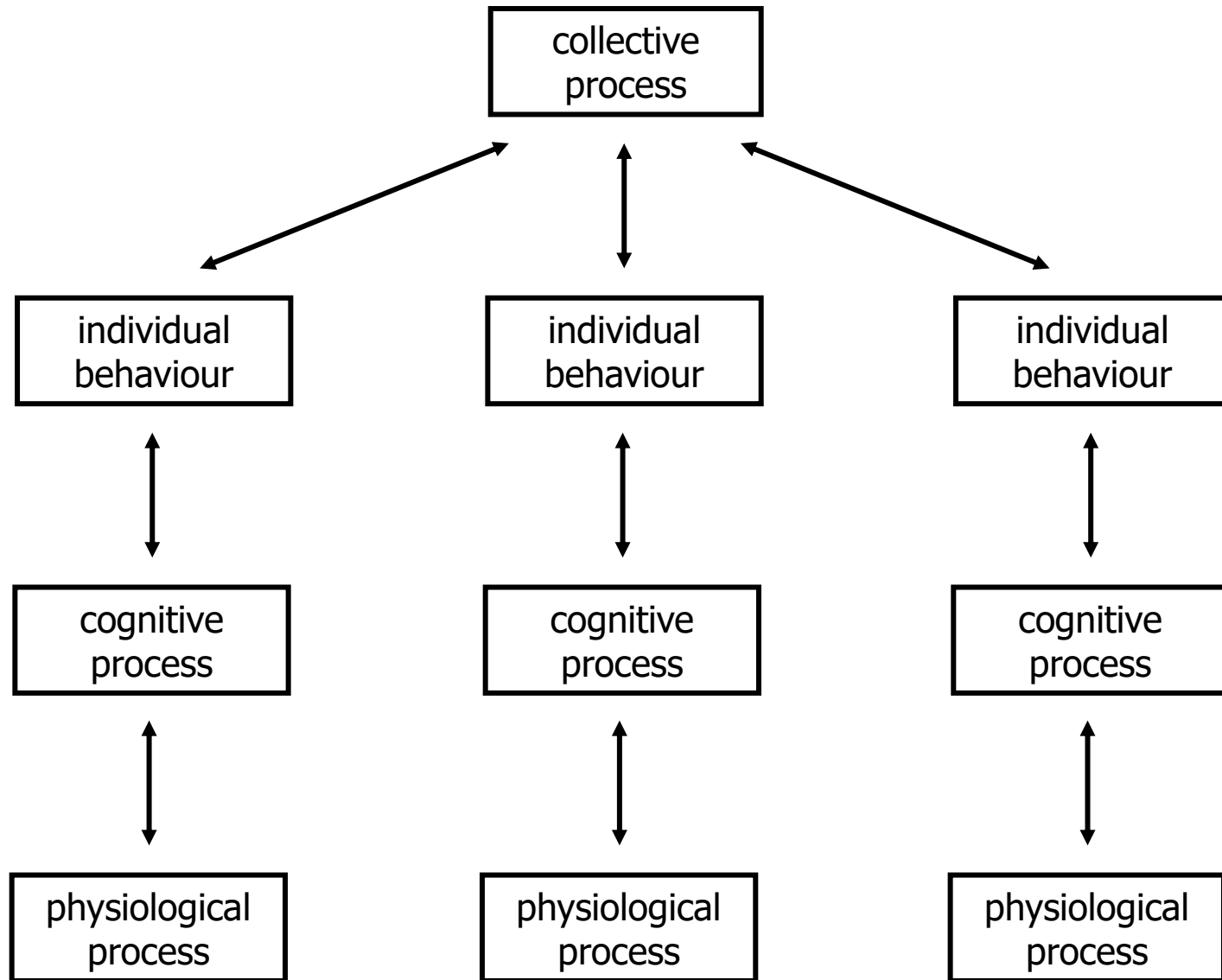
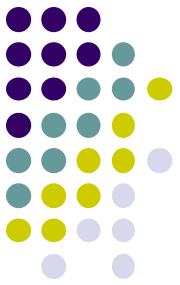
Modelling and Simulation Cycle

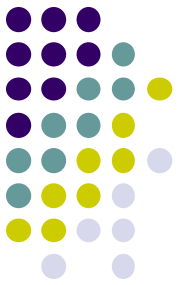




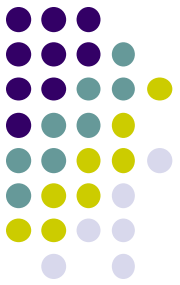
Steps

1. **Conceptualisation**: identification of important concepts and relations
2. **Formalisation**: describe concepts and relations in an unambiguous way
3. **Simulation**: running experiments with models to generate simulations traces
4. **Evaluation**: verify whether the simulations are in line with expectations (properties)



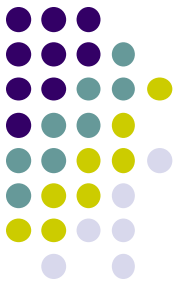


level	interlevel relations	content sciences
external collective societal level	<i>collective processes</i>	The diagram shows four levels of analysis in the first column, each with a corresponding process in the second column. Upward arrows indicate that higher-level processes emerge from lower-level ones. Brackets on the right group these levels into three perspectives: 'social perspective' (covering societal and individual levels), 'psychological perspective' (covering individual and cognitive levels), and 'biological perspective' (covering cognitive and physiological levels). The text 'emerging from' is placed between the arrows.
external individual behavioural level	<i>individual behaviours</i>	
internal cognitive level	<i>cognitive processes</i>	
internal physiological level	<i>physiological processes</i>	



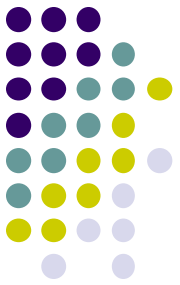
Numerical representation

- Concepts:
 - variables met numerical value
 - integer, real, 0 or 1
 - example: concept “temperature” → “Temp: real”
- Relations:
 - calculation rules to determine value of concept at next time point based on values other concepts
 - in dynamic model values change over time
 - Δt (change in time) is time step between two states



Logical representation

- Concepts:
 - statements (propositions) that can be *true* or *false*
 - example: `at_location(food, p3)`
- Relations:
 - Boolean rules that specify which combination of truth values is required for a concept to be true at next time point



(Semi-)formal notation

- Numerical temporal rules:

$$A(t + \Delta t) = A(t) + \alpha * ((C - B(t)) / C)$$

- **Semi-formal** logical temporal rules:

at any point in time,

if the animal observes that food is present at p2

and it observes that no screen is present,

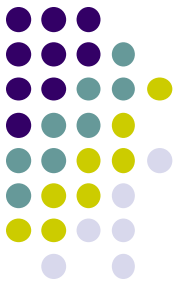
then it will go to position p2

- **Formal** logical temporal rules:

`observed(at_position(food, p2)) ∧ observed(no_screen)`



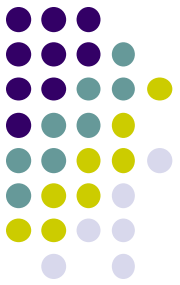
`performed(goto(p2))`



Learning goals

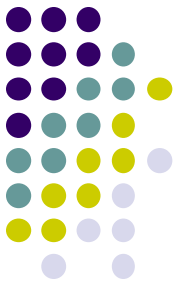
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Reporting

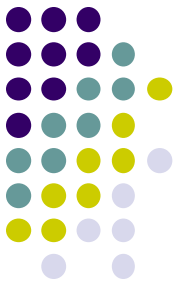
- Structured reporting is important
 - many arbitrary choices in process
- Elements of report:
 - assumptions, choices, motivation
 - concepts, relations, expected patterns
 - scenarios used in simulation experiments
 - settings
 - input parameters
 - Verified properties, conclusions



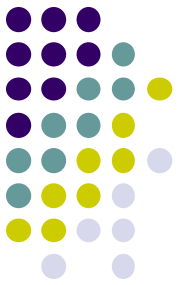
Assignment

Describe the characteristics of the worst report of a model (simulation) ever.

The worst report of a model (simulation) ever

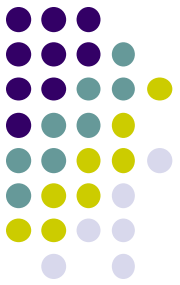


- No clear aim / scope / research question
- Formal concepts do not match informal concepts
- Formal relations do not match informal relations
- Specification does not run
- No evaluation
- No documentation of modelling choices
- Mix-up of different levels of aggregation
- Text is unreadable
- Handed in too late
- ...



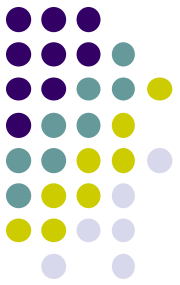
Building blocks for models

- Different models have been discussed
 - Bacterial Decision Making: Food uptake by E.Coli
 - Neural Mechanisms for Learning: Conditioning by Aplysia
 - Selforganisation in Honeybee Colonies
 - Epidemics from Contact Behaviour
- Models also served as illustrations for building blocks for other models



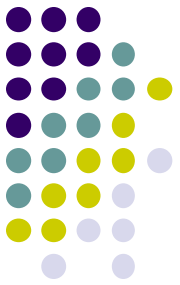
Building blocks

- Increase of factors
 - Growth:
 - linear
 - $Amount(t + \Delta t) = Amount(t) + c$
 - exponential
 - $Amount(t + \Delta t) = Amount(t) * \alpha$
 - logistic
 - $Amount(t + \Delta t) = Amount(t) * \alpha * ((C - Amount(t)) / C)$



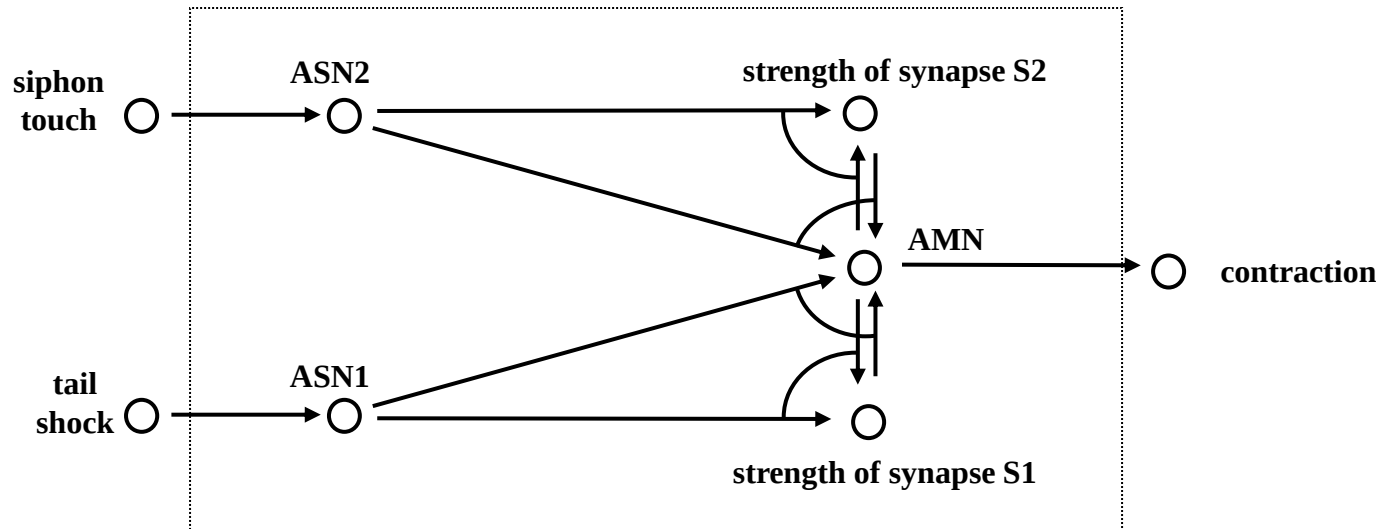
Building blocks – 2

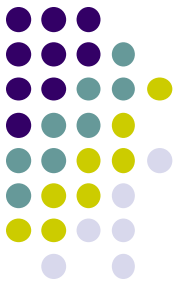
- Calculating with binary values in a numerical representation
 - AND
 - $C(t+\Delta t) = A(t) * B(t)$
 - NOT
 - $C(t+\Delta t) = (1 - B(t))$
 - OR
 - $C(t+\Delta t) = \text{MAX}(A(t), B(t))$



Building blocks – 3

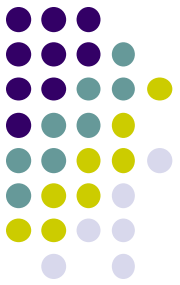
- Learning / adaptation:
 - Adjustment of the synapse strength based on simultaneous activation of MN and SN2
 - $S2(t+\Delta t) = \begin{matrix} S2(t) + 1 & \text{if } S2(t) < 4 \text{ and} \\ S2(t) & \text{AMN}(t) = \text{ASN2}(t) = 1 \\ & \text{else} \end{matrix}$





Building blocks – 4

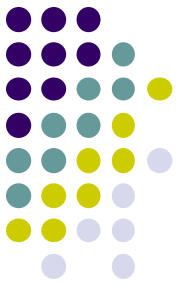
- Different types of behaviour:
 - reactive
 - stimulus – response
 - delayed response
 - motivation-based
 - via desires and intentions
 - adaptive
 - via “sensitivity”
- Environmental Complexity Thesis
 - *The more **complex the environment** → the **more sophisticated the behaviour** that is required to deal with this environment → the **more complex the mental representations** and capabilities needed.*



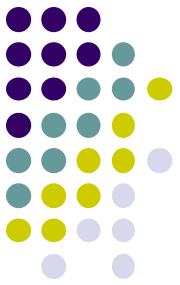
Building blocks – 5

- Vocabulary for behavioural & agent modelling
 - external perspective:
 - observations & actions
 - internal perspective:
 - cognitions
 - beliefs, sensory_representations, preparation_states
 - extented mind
 - no cognitions, only observations + actions
 - agent
 - internal concepts: world model, self model, history
 - desires, intentions, communications,

Common agent concept formalisation

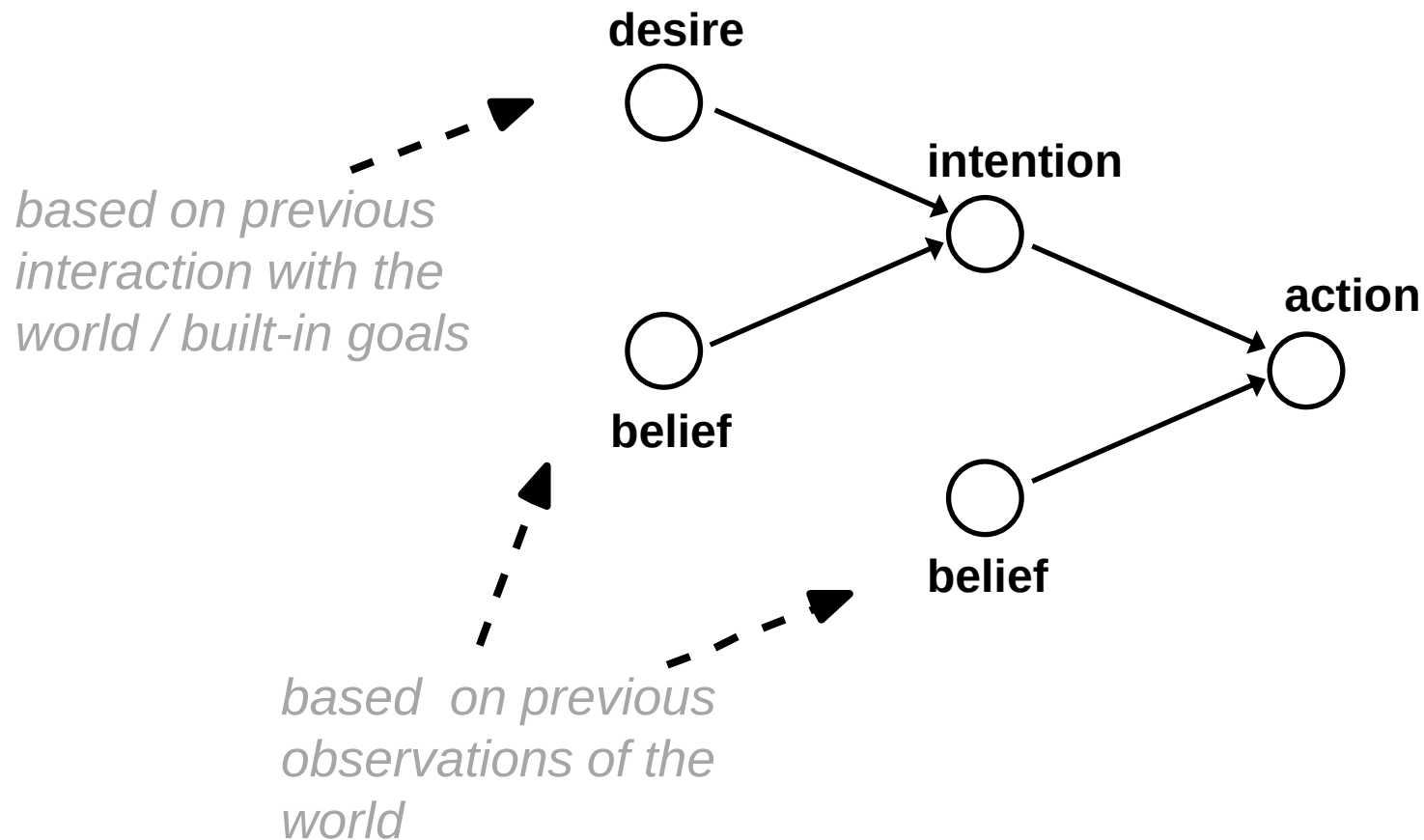


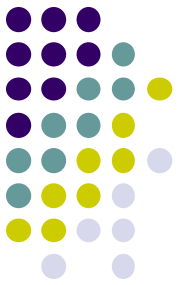
<i>concept</i>	<i>predicate</i>
sensory_represenation	sr : AGENT x OBSERVATION
preparation_state	ps : AGENT x ACTION
belief	belief : AGENT x INFO_ELEMENT
desire	desire : AGENT x INFO_ELEMENT
passive observation	observed : AGENT x INFO_ELEMENT
active observation	to_be_observed : AGENT x INFO_ELEMENT
action	performed : AGENT x ACTION
incoming communication	communicated_from_to : INFO_ELEMENT x AGENT x AGENT
outgoing communication	to_be_communicated_from_to : INFO_ELEMENT x AGENT x AGENT



Building blocks – 5

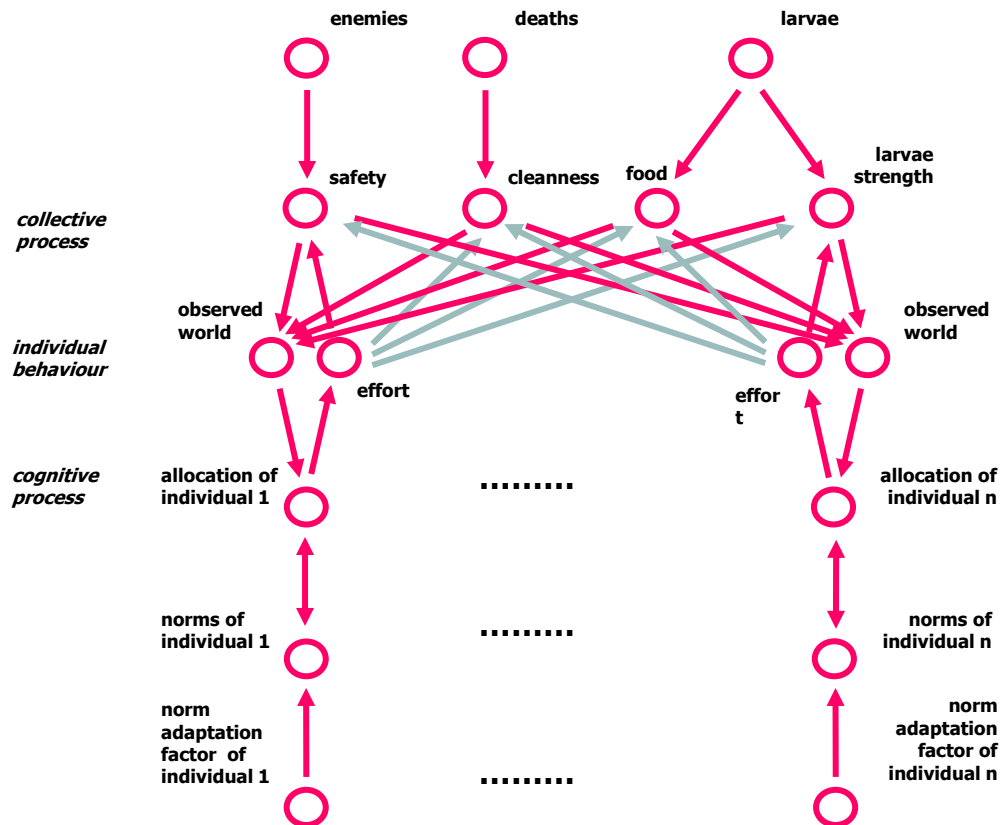
- BDI model:

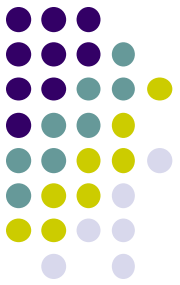




Building blocks – 6

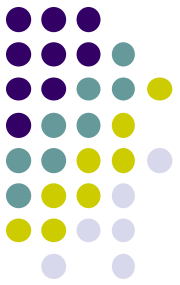
- Integration of models of different levels
 - cognition – behaviour – collective





Building blocks – 7

- Two approaches:
 - Modeling **groups**
 - population-based (collective)
 - one concept for number of individuals in a category
 - model describes **transitions** between categories
 - *abstracted but scalable*
 - Modeling individuals
 - individual-based
 - one model per individual, individual behaviour
 - *precise but not easy to scale*



Building blocks – 8

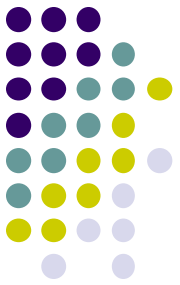
- Population-based models usually require stochastic variables
 - precise factors have been abstracted

InfectionStateA1(t+Δt) = 1 IF

InfectionStateA1(t) = 1 AND $r \geq \text{RecoveryRate}$

(where r is random number between 0 and 1)

- Stochastic models require several simulation runs to draw conclusions

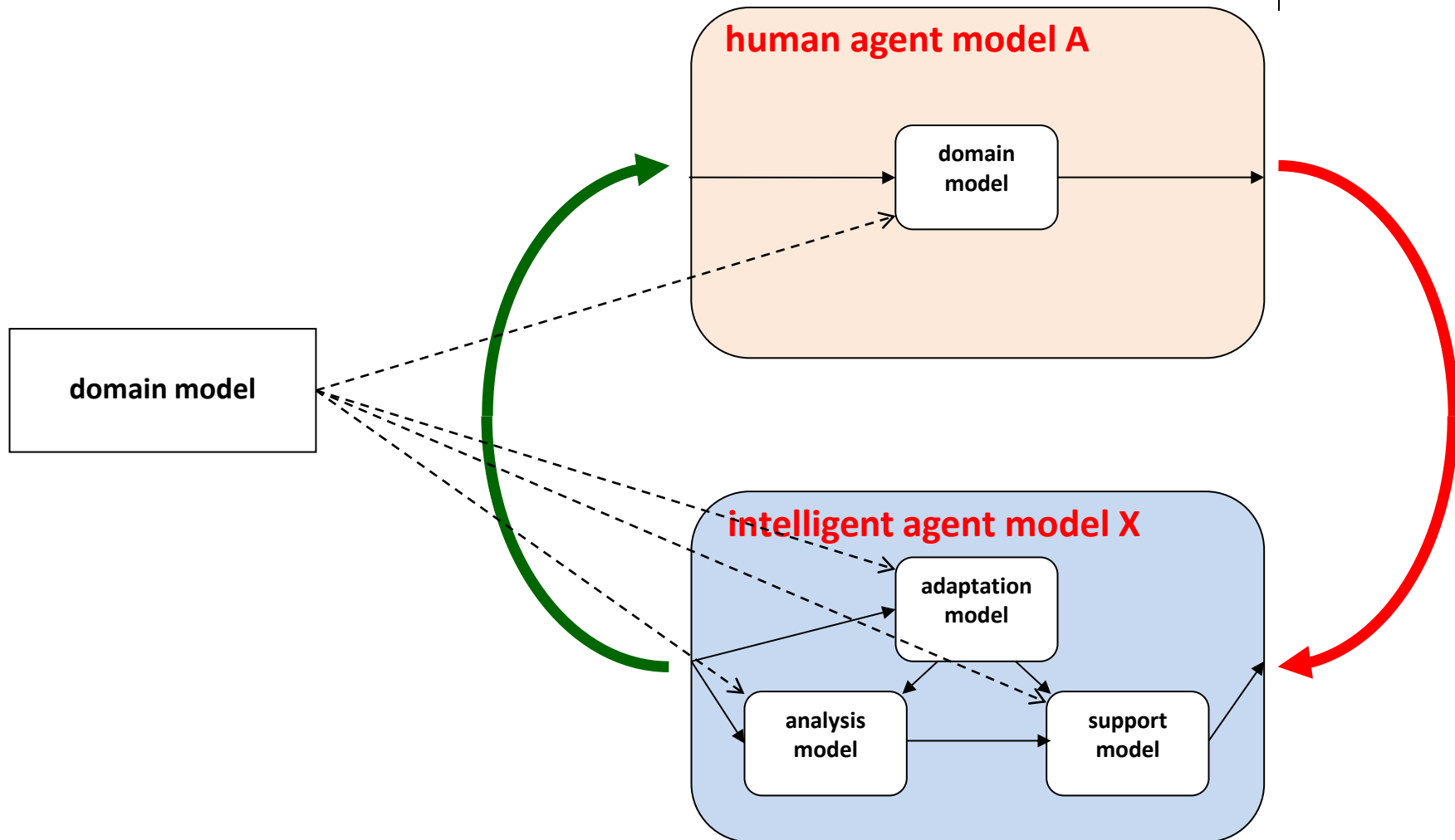
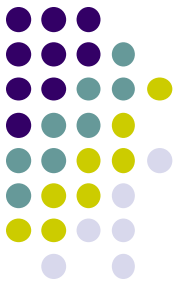


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Four ways of using a domain model in intelligent applications



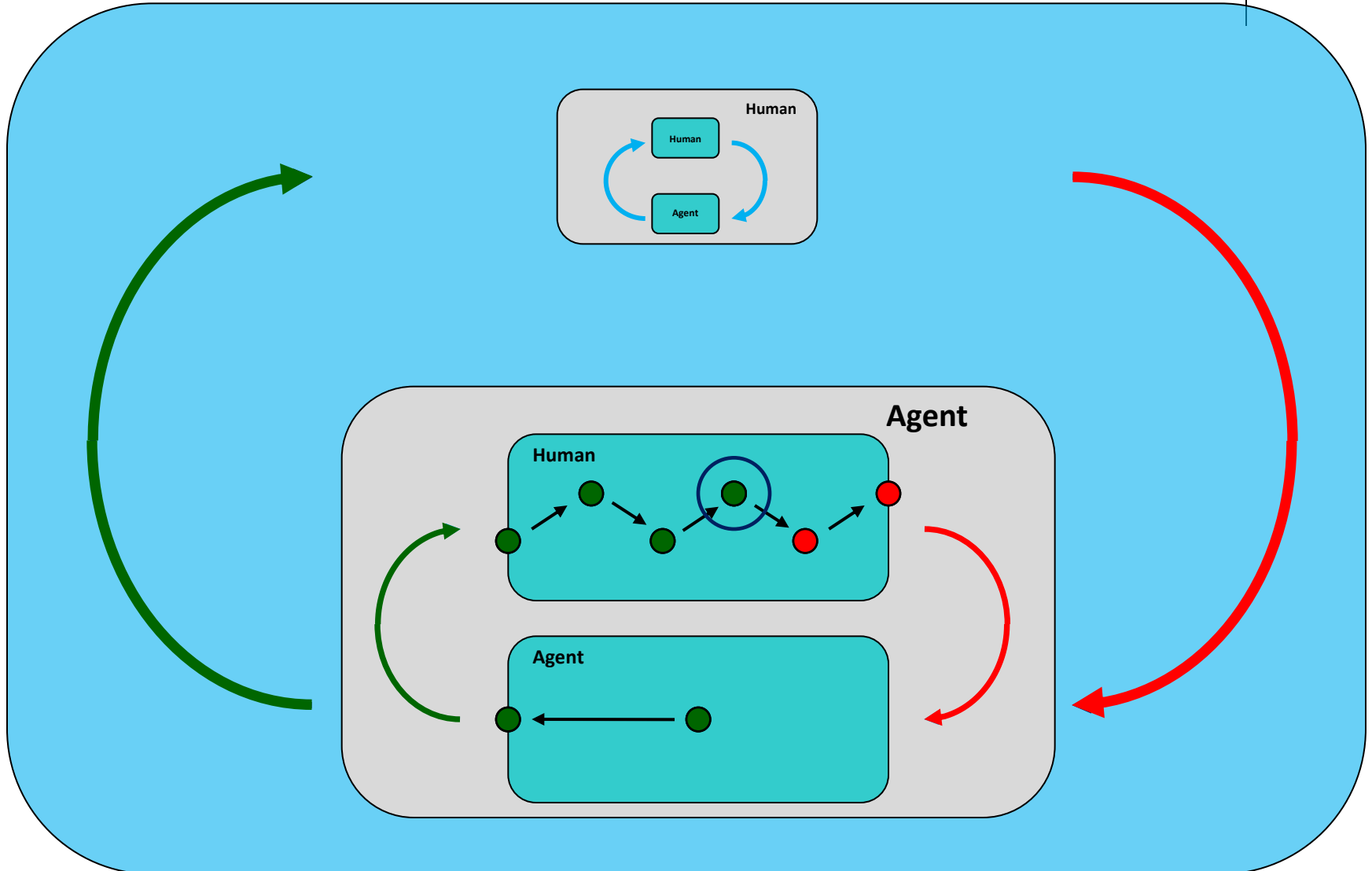
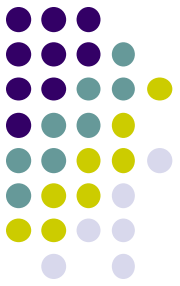


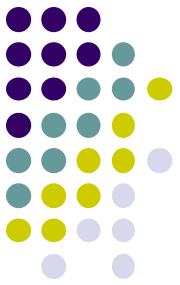
Integration: Four ways

- **domain** model ←
 - represents human process
- **analysis** model
 - to analyse and assess the process
- **support** model
 - to generate support for the process
- **(parameter adaptation** model)
 - (to improve the model of the process)

preferably
based on
scientific
theory!

Human Models within Agents



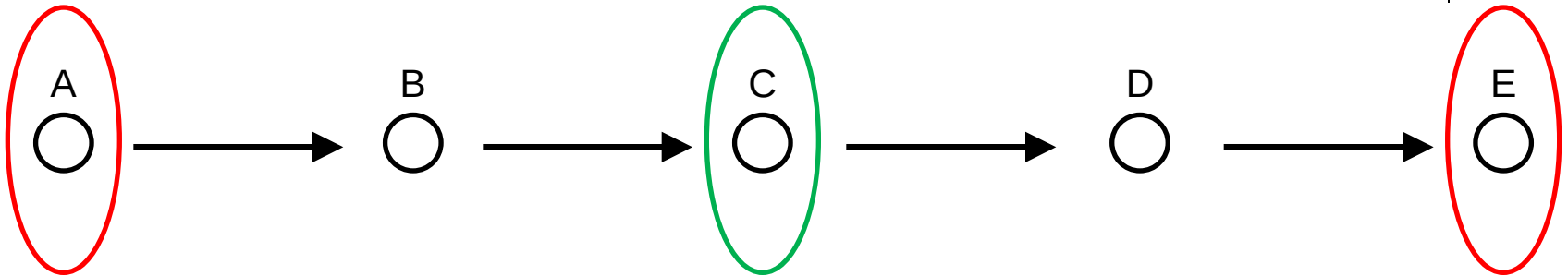


Embedding domain models

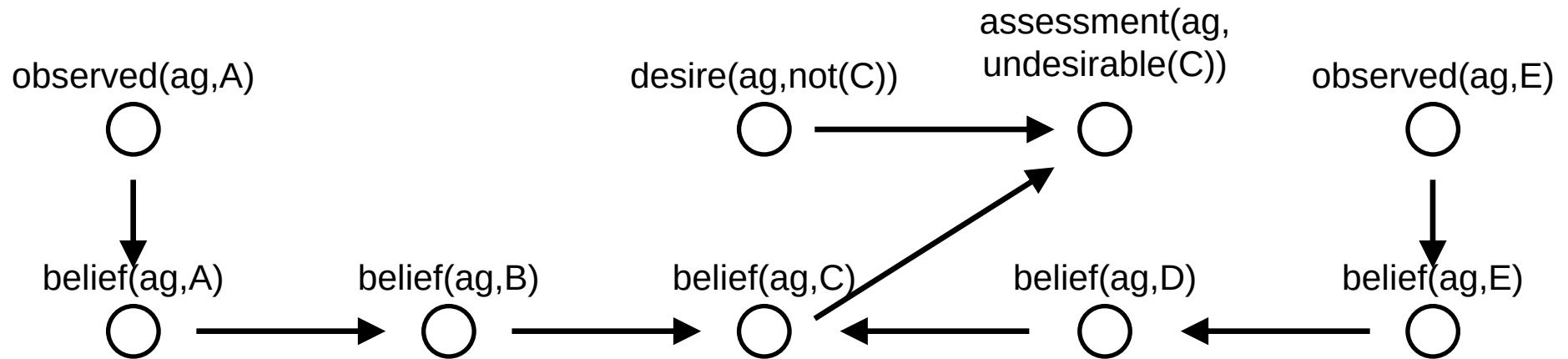
- Domain model → Analysis model:
 - Use meta-statements about domain model
 - Reason from **observations** and **desire** to **assessment**
 - Forward or backward reasoning (or both)
- Domain model → Support model:
 - Extend domain model with support actions
 - Backward reasoning: from **desired state** to **support action**
 - Reasoning about **desires**
 - Forward reasoning: from **assumed action** to **desired effect**
 - Reasoning about (hypothetical) **beliefs**



Domain Model $\rightarrow$ Analysis Model

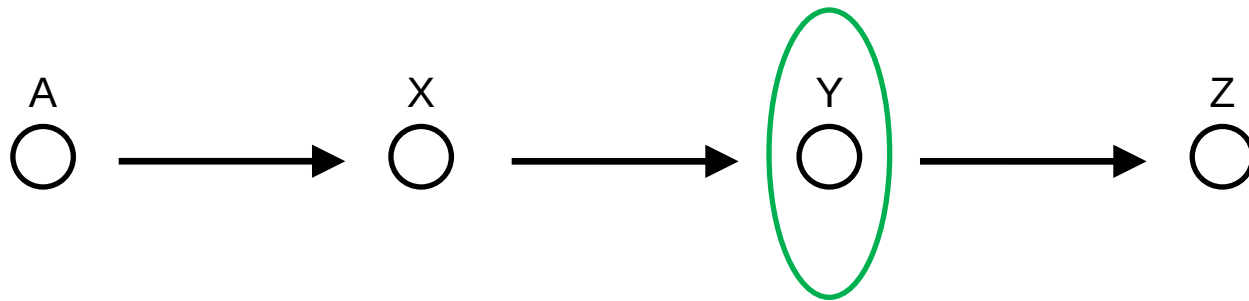
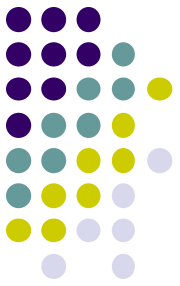


DOMAIN

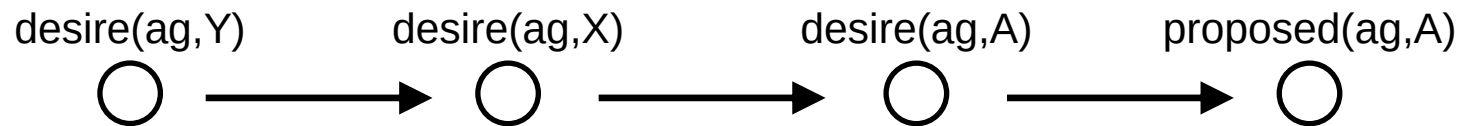


ANALYSIS

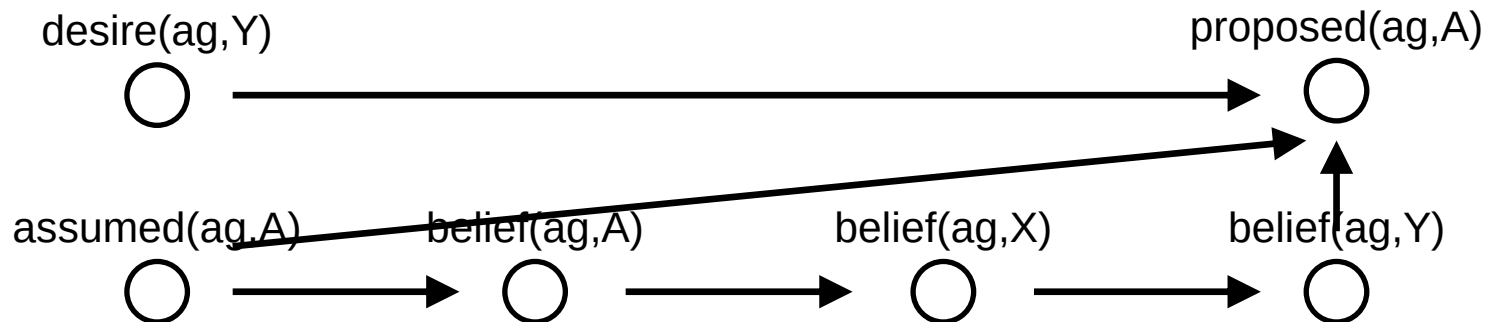
Domain model → Support model



DOMAIN



SUPPORT (backward)



SUPPORT (forward)

Question time!

